PROJECT AND TEAM INFORMATION

VeriNews is the model we propose to solve the problem.

Project Title

VeriNews: An Explainable Multi-Factor News Verification and Credibility Scoring Engine Using DSA in C/C++

Student / Team Information

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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

In the current digital environment, false information and misleading headlines can become viral via social media, messaging services, and online news sites. Such information has the power to influence public opinion, cause unwarranted fear, and damage faith in reliable journalism. Since the amount of online content is still increasing, it is becoming more and more difficult for people, particularly students, to decide quickly whether or not a news headline or article is trustworthy.

A large number of the fact-checking systems in use today rely very much on manual verification, a process which can be extremely time-consuming. Some, on the other hand, make use of sophisticated machine learning models that generally do not provide any clarity regarding the way in which they reach their conclusions. This shows that it is necessary not only to decide whether or not a news item is fake or genuine but also to give users a clear and easy-to-understand explanation of its credibility.

In order to deal with this problem, we are developing **VeriNews**, a fast and explainable news verification system based on basic data structures in C and employing object-oriented programming principles in C++. The system identifies key entities and keywords from news articles and checks them against information in curated databases which contain both verified and unverified entries. It then calculates a credibility score taking into account factors such as the reliability of the source, the relevance of the topic, and time consistency. The system seeks to provide users with a simple, informative, and transparent method of evaluating the credibility of online news by combining efficient computational techniques with an explainable verification process, thus enabling them to make more informed decisions.

State of the Art / Current solution (1 pt)

Currently, fake news verification is mostly limited to two main approaches. One is manual fact-checking by experts, as seen in platforms like Snopes, Alt News, and PolitiFact.These are slow but accurate.On the other hand, automated systems use algorithms to detect false news quickly.However, they often lack the ability to understand context and nuances.As a result, users are forced to choose between slow but accurate methods or fast but unreliable ones.There is very little in between.Many systems also rely on cloud-based AI or NLP APIs, which are not transparent.Manual methods take too long, allowing fake news to spread before it can be corrected. On the technology front, most platforms use complex neural networks or proprietary cloud-based fact-checking tools.

These systems work like black boxes, giving users no insight into how they make decisions or how they evaluate sources.Also, existing systems rarely provide simple, standalone rule-based software that can run locally with efficient algorithms, without needing many external dependencies.

Project Goals and Milestones (2 pts)

Our goal is to build a high-performance verification and credibility-scoring application that allows users to submit news headlines or article snippets and immediately receive a weighted credibility score (0–100%) along with a detailed explanation.

Project Milestones:

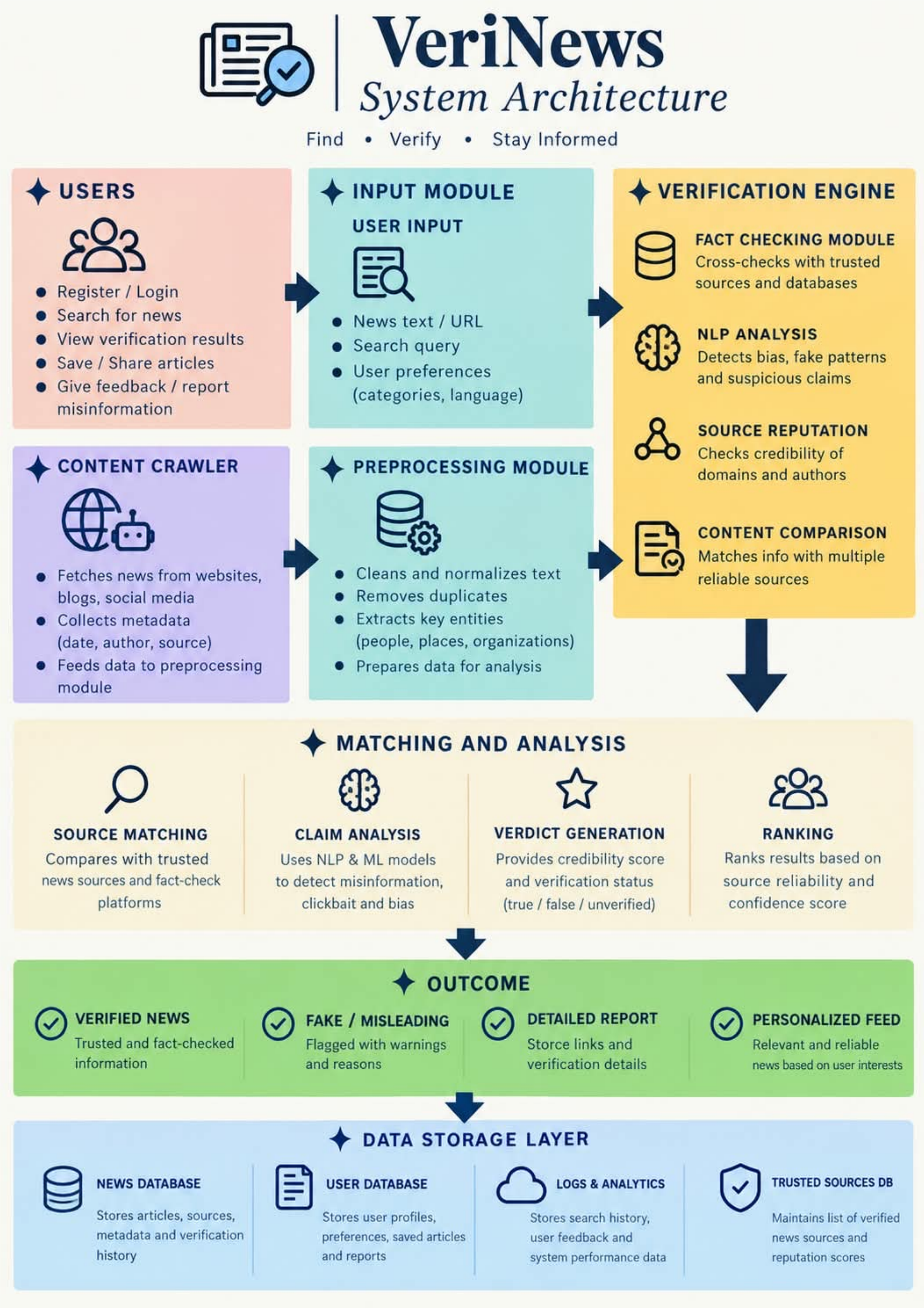
- **Milestone 1 (Phase 1):** Implement custom C data structures: Hash Tables (indexing), Linked Lists (records), and BSTs (sorting/filtering).
- **Milestone 2 (Phase 2):** Build string tokenization, stop-word filtering, entity extraction, and the weightedcredibility scoring engine.
- **Milestone 3 (Phase 2/3):** Implement C++ OOP architecture (User, News,Admin, Engine),a FIFOrequest queue, and anundo/history stack.
- **Milestone 4 (Phase 3):** Build file I/O for datapersistence and theinteractiveAdmin CRUD module.
- **Milestone 5 (Phase 3):** Conduct unit testing, complexity profiling, and compile final submission deliverables.

Project Approach (3 pts)

The system combines performance-oriented data structures with an object-oriented C++ control layer.

- **Text Processing & Keyword Extraction:** Cleaninput text, removeunnecessaryelements, filter stop words, and extract key entities.
- **Data Structures & Matching Layer (C):** HashTablesareusedforquickkeywordlookupsinnearconstanttime,dynamicLinkedListsareused to chain records, Binary Search Trees (BST) are used for filtering based on time or score ranges, FIFO Queues are used for grouping input data, and Stacksareused to track operation history.
- **Multi-Factor Verification Engine:** This system calculates a score based on several factors: Entity Match (40%) Source Reliability (25%) Corroborating Reports (20%) Temporal Consistency (10%) and Category Alignment (5%).
- **Classification & Persistence (C++ OOP):** Output is categorized into three groups: Likely Verified (80–100%) Needs Verification (%) or Suspicious (0–49%).

System Architecture (High Level Diagram) (2 pts)



Project Outcome / Deliverables (1 pt)

By the end of this project, we will deliver a fully working, standalone C/C++ application called **VeriNews** that allows users to assess news headlines and view detailed credibility scores.
Key Deliverables include:

- Source Code: Fully modular C (custom ADTs) and C++ (classes and control flow) code.
- Dataset & Persistence Layer: Seeded news corpus, trusted source registry, and file-based local storage.
- Admin Module: Interactive console interface for full database CRUD operations.
- Documentation: Final project report with Big-O complexity analysis and UML diagrams.

Assumptions

Input text is English and provides sufficient named entities for tokenization. Evaluations rely on a curated local truth dataset; it is an offline verification engine, not a live web crawler.

References

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